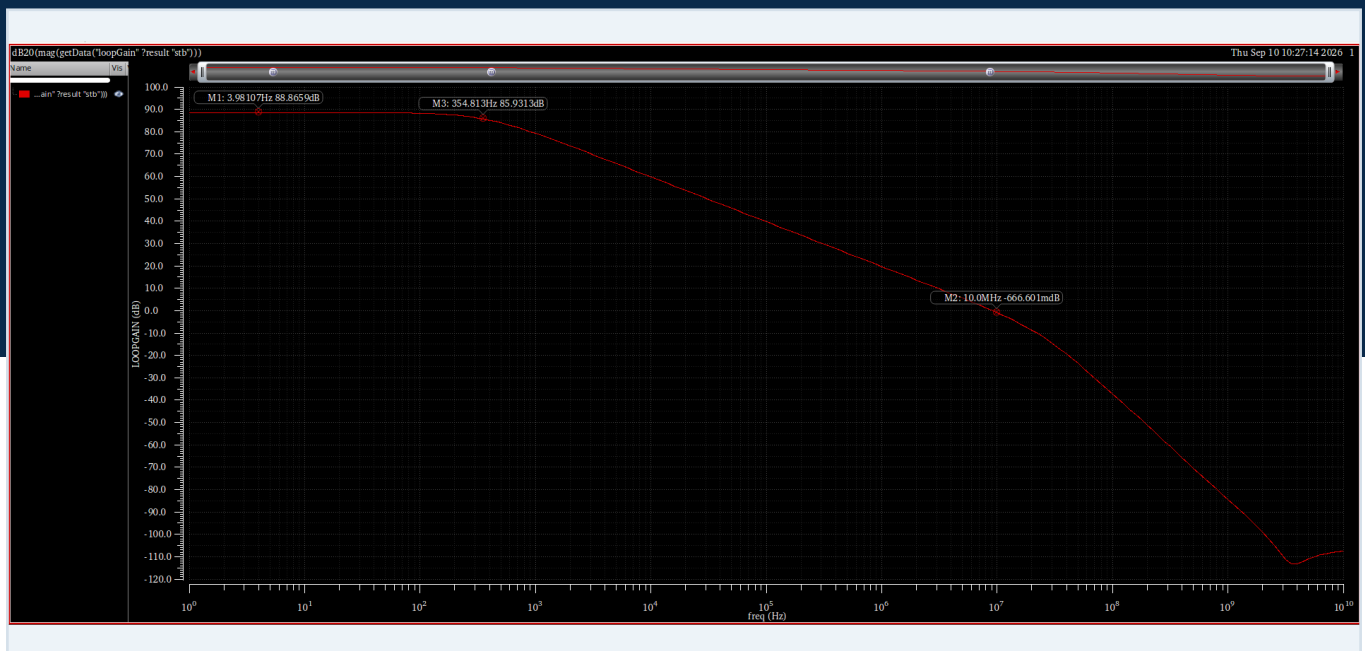


Rail-to-rail input. Monticelli biasing. Class-AB output.

A 65 nm operational amplifier: nominal loop closure, common-mode sweeps, load drive, and large-signal response.

NOMINAL SIMULATION VERIFICATION



Selected simulation evidence. Full conditions and interpretation appear in the report.

LATEST NOMINAL GBW

10.11 MHz

LATEST NOMINAL PM

60.58 deg

SUPPLY

2.5 V

Compiled from the author's original report, screenshots, and saved simulation records. Simulations were not rerun for this edition.

A complete nominal evidence trail.

This design combines complementary rail-to-rail input stages, a folded-cascode signal path, Monticelli translinear-loop biasing, and a class-AB output stage. The archived nominal stability run narrowly clears the specified gain-bandwidth and phase-margin targets.

Metric	Latest nominal record	Requirement
Gain-bandwidth product	10.1083 MHz	Greater than 10 MHz
Phase margin	60.5839 degrees	Greater than 60 degrees
Low-frequency loop gain	~88.9 dB, separate captured plot	Greater than 60 dB
Nominal input common mode	1.25 V	Midrail test
Compensation variable c	1.65 pF	Tuned for stability
Simulation errors	0	Successful nominal run

TEST-BENCH BASELINE

Condition	Assignment setup
Technology / models	65 nm, 2.5 V IO MOS devices; TT / 27 C
Supply / midrail	VDD = 2.5 V; VMID = 1.25 V
Load	RL = 10 kohm to midrail; CL = 20 pF
Output-stage bias target	200 uA
Closed-loop configurations	TB1: unity buffer; TB2: gain +2; TB3: gain -1

Result scope

The assembled report is complete as a reconstruction of the available evidence. The design status remains nominal verification: full-range GBW, PVT, mismatch, and post-layout closure are not established.

Complementary input to class-AB drive.

Complementary input pairs extend common-mode coverage. Their signal currents pass through the folded signal path into the output-control nodes. A Monticelli translinear loop establishes the class-AB bias relationship, allowing both output devices to conduct at quiescence and to supply larger load currents when required.

Block	Role in the implemented amplifier
Complementary input pairs	Maintain an input signal path as common mode approaches either supply.
Folded-cascode signal path	Combines and conveys the input-stage currents to high-gain nodes.
Monticelli bias network	Controls the complementary output-stage quiescent conduction.
Class-AB output stage	Sources and sinks load current around a midrail operating point.
Compensation / load	Small compensation capacitors plus the specified 20 pF load.

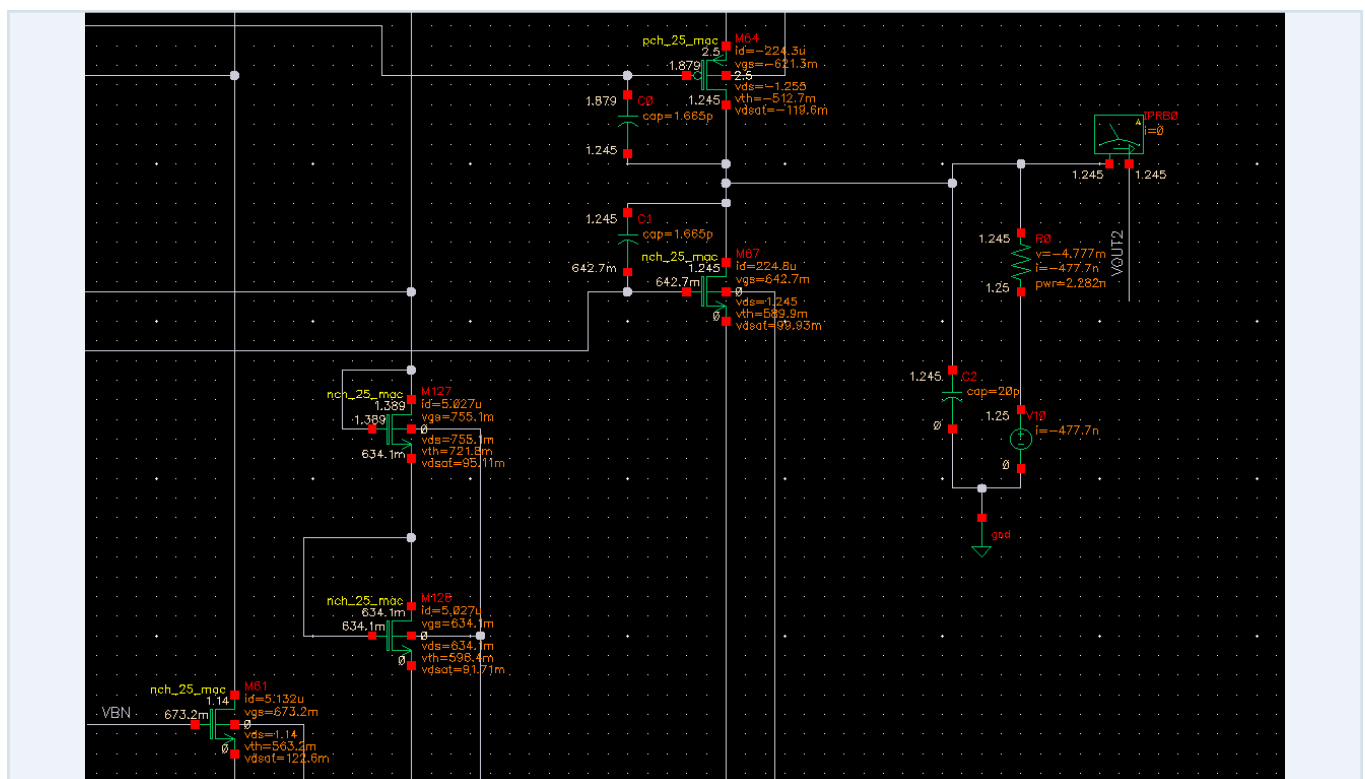


Figure 1. Annotated output stage and load. The implemented devices are M64 (PMOS) and M67 (NMOS). V_{OUT} is approximately 1.245 V.

The archive preserves several close-up operating-point views rather than one complete, high-resolution schematic. The report uses those views directly and avoids inferring a full transistor connectivity or sizing table from a thumbnail.

Quiescent conduction and bias checks.

Saved operating-point quantity	Approximate annotation
PMOS output-device M64 current magnitude	224.3 μ A
NMOS output-device M67 current	224.8 μ A
Output voltage	1.245 V
Load midpoint	1.25 V
Output-stage bias target in brief	200 μ A
Nominal compensation variable, final run	c = 1.65 pF

Both complementary output devices carry current at quiescence, consistent with the intended class-AB bias. The captured currents are about 12% above the stated 200 μ A bias target. The small difference between source and sink current is consistent with the load's slight offset from midrail.

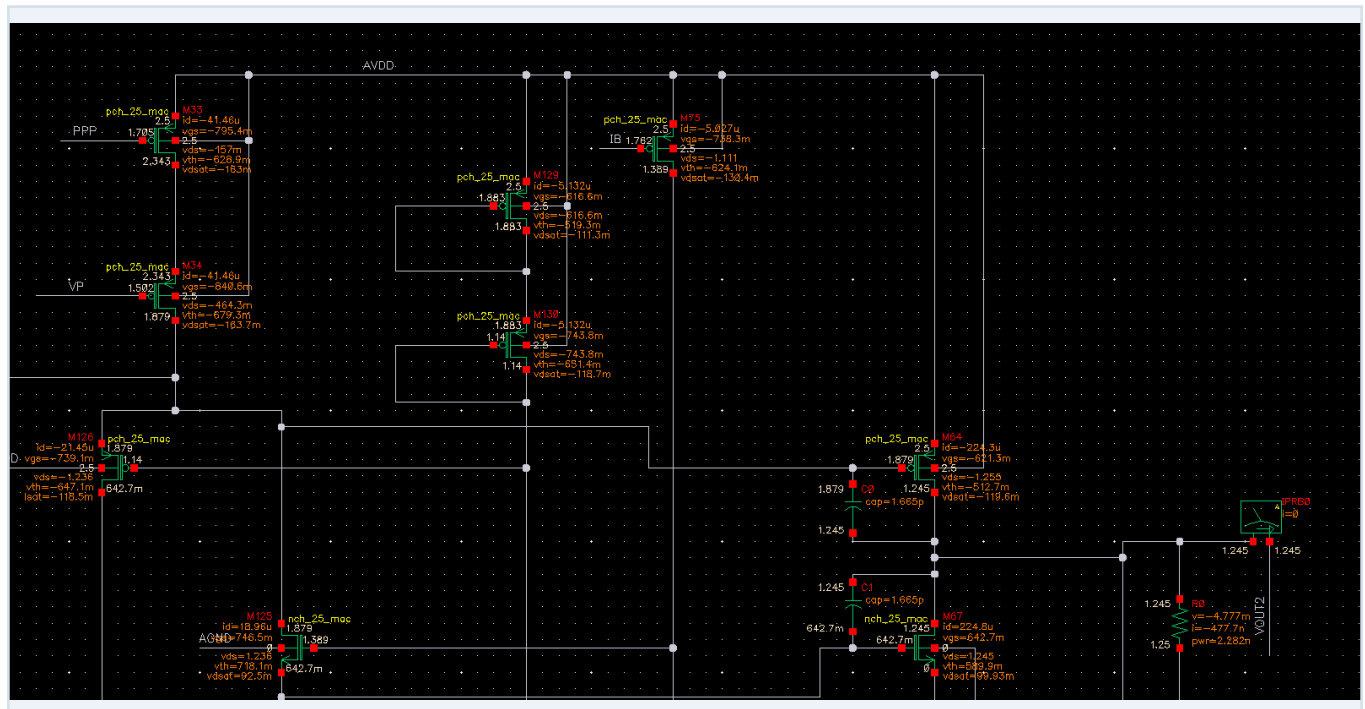


Figure 2. Additional annotated local operating-point view from the implemented amplifier. Original device names and annotations are retained.

Nominal bandwidth and phase margin.

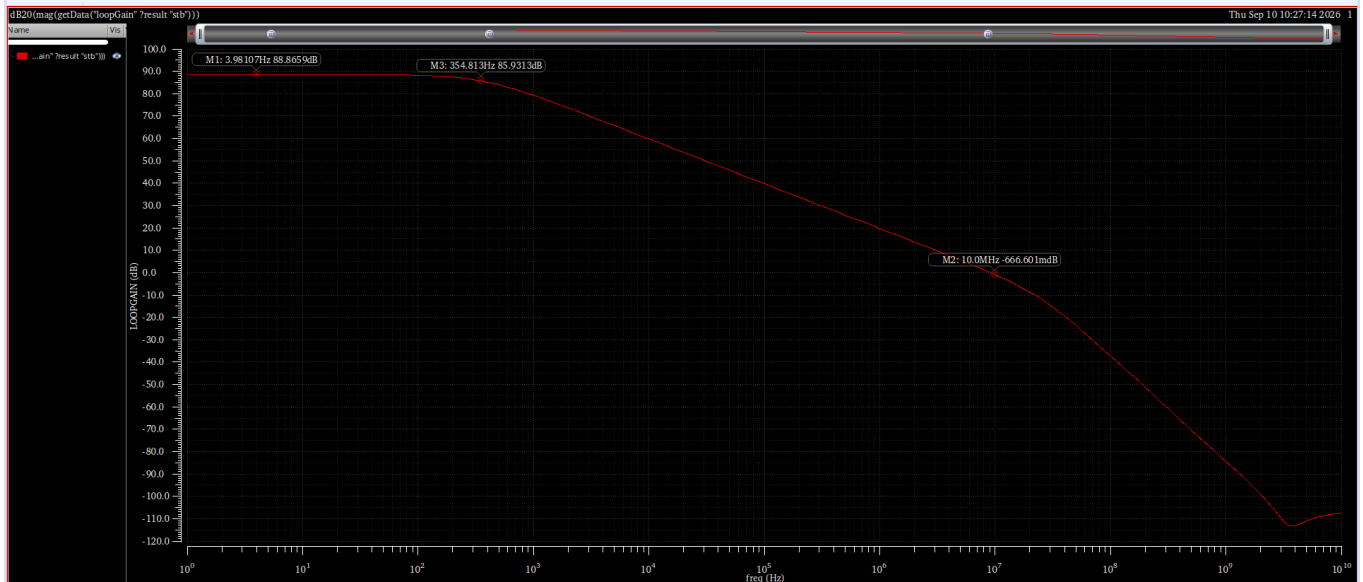


Figure 3. Captured loop-gain response. This plot is a separate archived snapshot from the final numerical run.

Captured plot marker	Value
Low-frequency loop-gain magnitude	88.8659 dB at 3.98107 Hz
Near -3 dB relative to low-frequency gain	85.9313 dB at 354.813 Hz
Loop magnitude at 10 MHz	-0.666601 dB

Saved run	GBW	PM	Other
Screenshot 172628	10.03 MHz	60.21 deg	GM = 17.24 dB
Interactive.184 final	10.1083 MHz	60.5839 deg	0 errors

Interactive.184 uses $V_{CM} = 1.25$ V and $c = 1.65$ pF at nominal $T_T / 27$ C. Its margin is about 0.108 MHz above the 10 MHz GBW target and 0.584 degrees above the 60-degree PM target. The gain margin belongs to the screenshot run and is not attributed to the final record.

Common-mode coverage and stability.

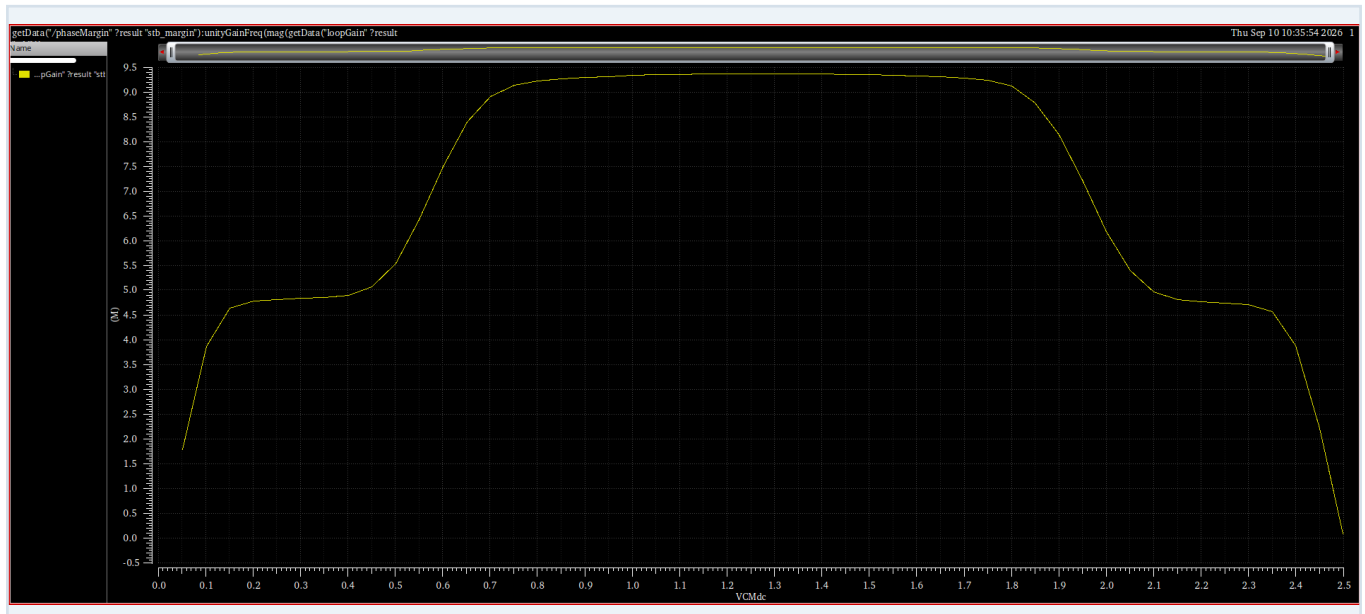


Figure 4. Unity-gain frequency, f_u , versus input common-mode voltage. This is not gain-bandwidth product.

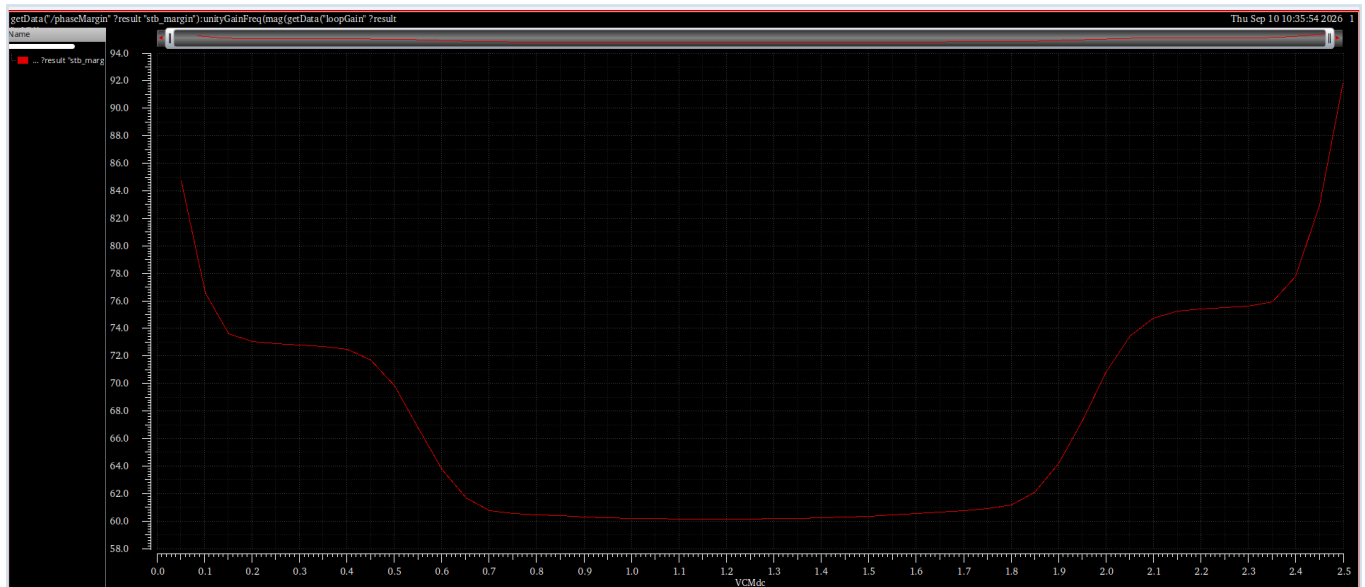


Figure 5. Phase margin versus common-mode voltage for the captured sweep; the visible minimum is approximately 60.1 to 60.2 degrees.

The f_u curve has a broad midrange plateau near 9.2 to 9.4 MHz, approximately 9.37 MHz at midrail, and falls near either rail. The PM curve has its smallest visible margin near midrail and rises toward the rails.

Two measurements, two meanings

`gainBwProd()` and `unityGainFreq()` are different measurements. A 10.1083 MHz nominal GBW value must not be substituted for the f_u sweep, nor used to claim greater than 10 MHz bandwidth over the entire rail-to-rail input range.

DC transfer into the specified load.

The unity-buffer DC sweep varies the input from about 0.2 to 2.3 V. With a 10 kohm load connected to 1.25 V, the output follows the input while the complementary output-device currents redistribute.



Figure 6. DC output response under the 10 kohm load.

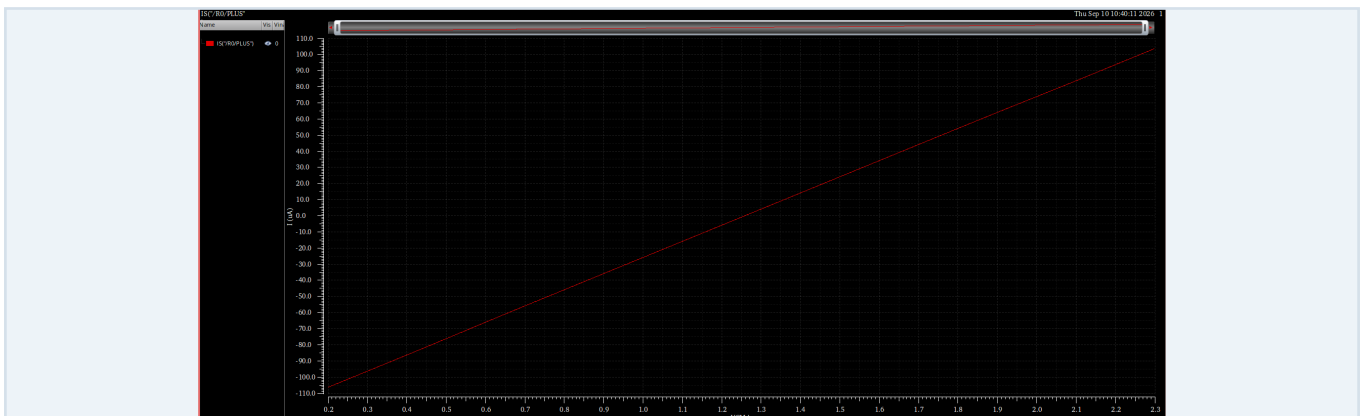


Figure 7. Load current during the 10 kohm DC sweep.

Plot-derived quantity	Approximate visible range
Output voltage	0.205 to 2.282 V
PMOS M64 current magnitude	192 to 277 uA
NMOS M67 current	298 to 173 uA as input rises
Load current	-105 to +103 uA

A heavier load exercises class-AB drive.

A second DC sweep reduces the load to 2 kohm while retaining the same midrail reference. The output-stage currents change more strongly, illustrating the current-drive capability of the class-AB arrangement.



Figure 8. DC output response under the 2 kohm load stress test.

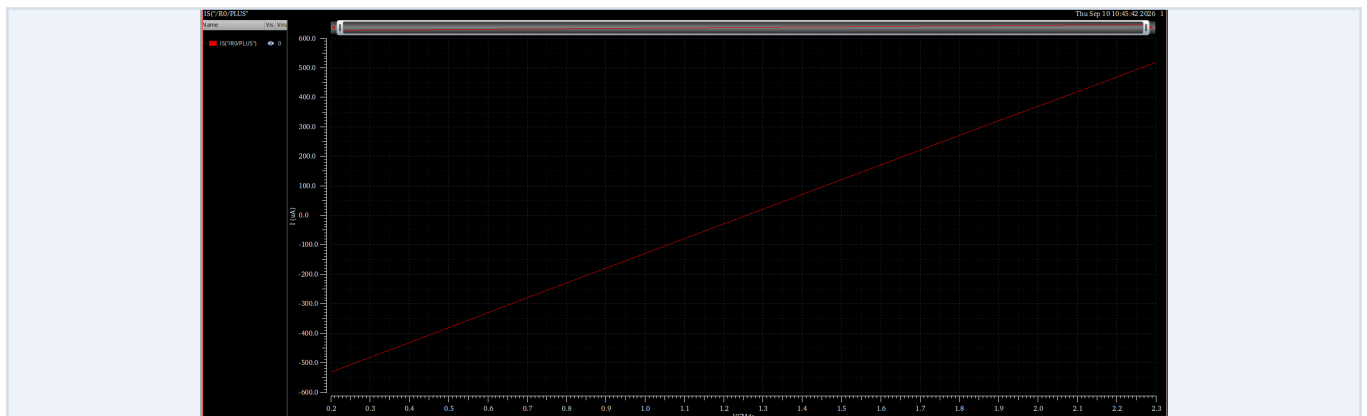


Figure 9. Load current approaches roughly half a millipere in either direction.

Plot-derived quantity	Approximate visible range
Output voltage	0.205 to 2.282 V
PMOS M64 current magnitude	157 to 651 uA
NMOS M67 current	687 to 132 uA as input rises
Load current	-531 to +518 uA

Positive slew rate from saved cursors.

The unity-buffer transient applies an approximately 0.75 to 1.75 V input step. Saved cursor annotations provide two measurements on a positive-going transition.

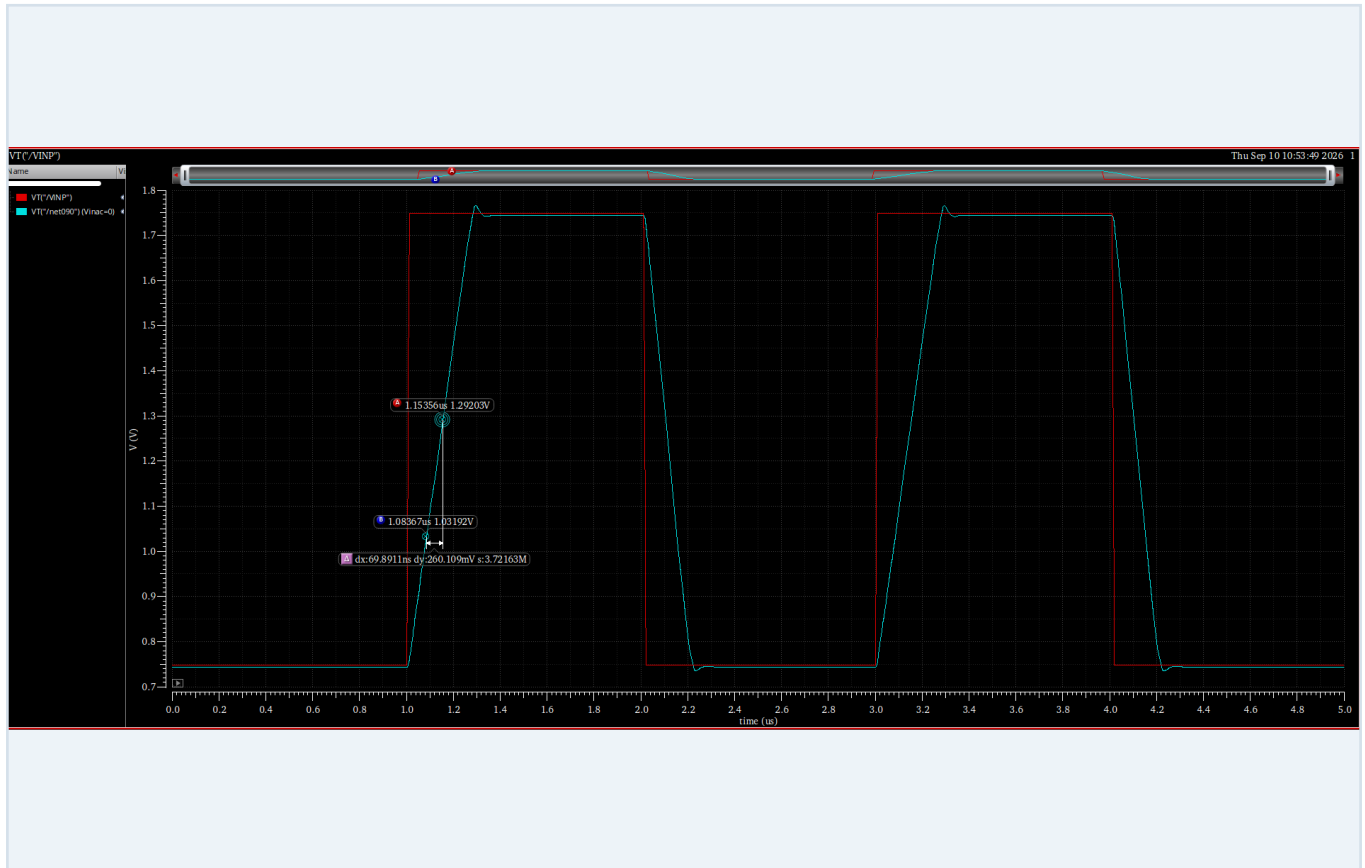


Figure 10. Unity-buffer step response with a positive-slope measurement of approximately 3.72 V/us.

Cursor capture	delta-V / delta-t	Positive slope
Screenshot 175435	315.893 mV / 85.2018 ns	3.70759 V/us
Screenshot 175449	260.109 mV / 69.8911 ns	3.72163 V/us

A concise source-backed result is SR+ approximately 3.72 V/us. Both numerical cursor readings are positive slopes; they must not be presented as separate positive and negative slew rates.

Remaining measurement

The archive does not contain a saved numerical SR- value or a cursor-based settling time for this step. These should be measured explicitly before reporting a symmetric slew or settling specification.

Non-inverting and inverting operation.

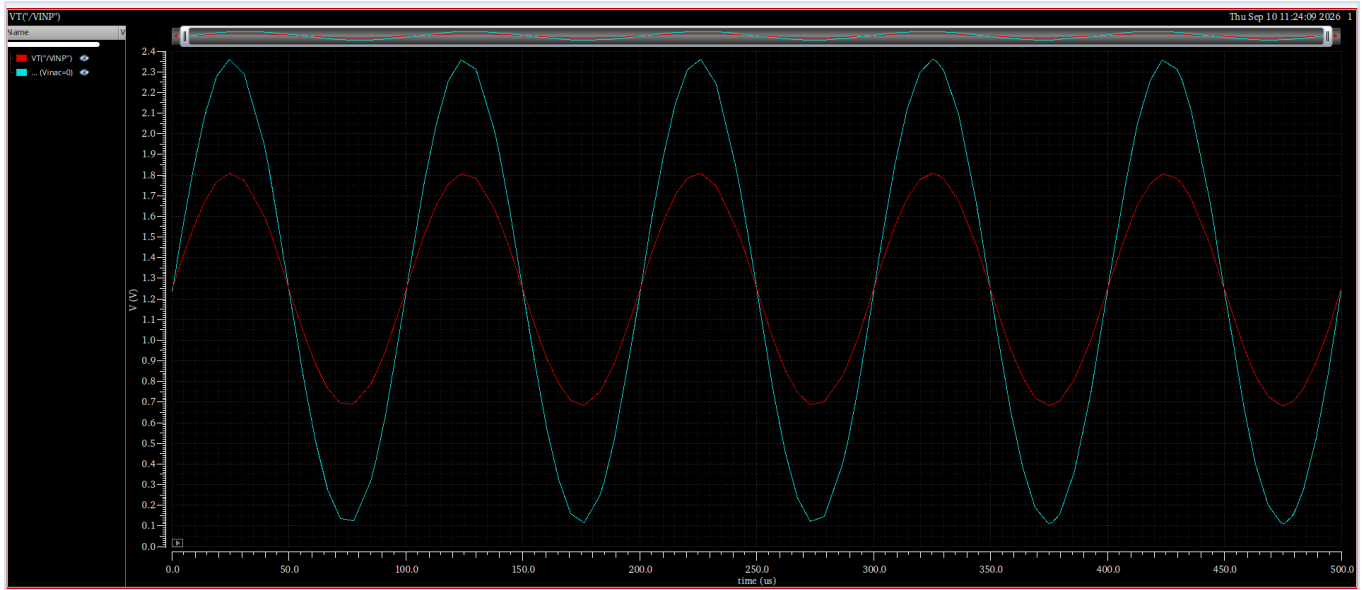


Figure 11. TB2, non-inverting gain +2, at 10 kHz. Approximate input span: 0.69 to 1.80 V; output: 0.11 to 2.36 V.

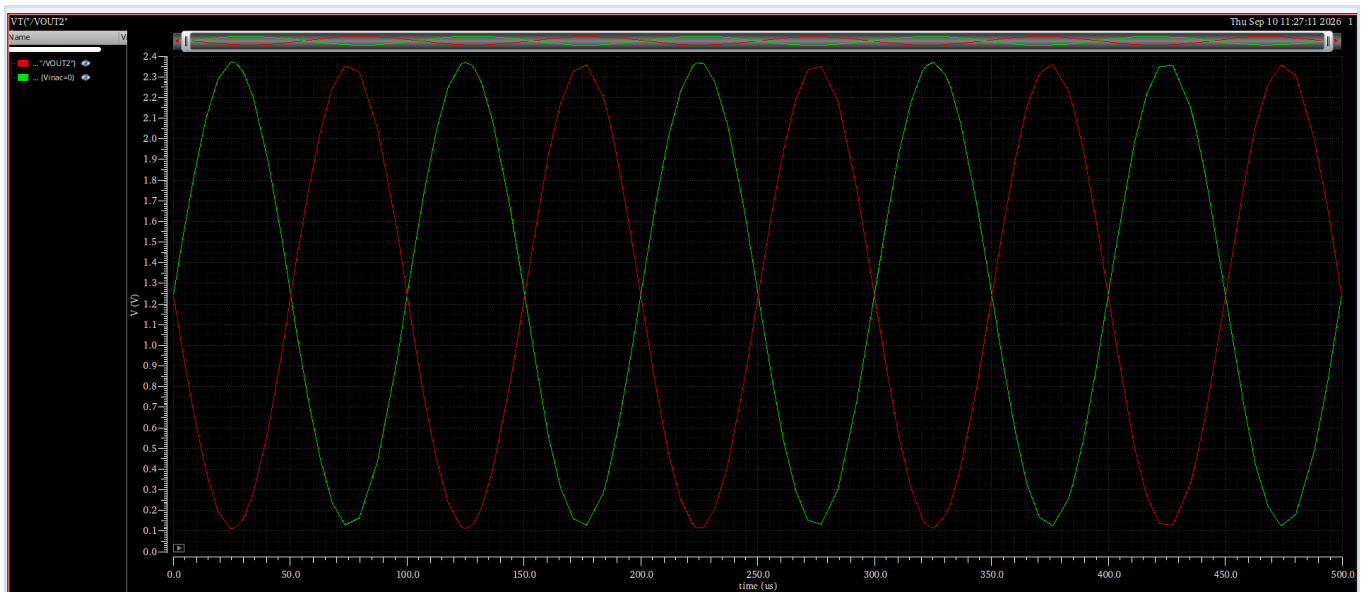


Figure 12. TB3, inverting gain -1, at 10 kHz. Input and output are inverted and span approximately 0.12 to 2.36 V.

The saved waveforms show an output excursion of roughly 2.24 to 2.25 V peak-to-peak. These axis-derived values support a broad large-signal swing demonstration in the specified configurations.

Precision of the swing claim

No exact full-swing cursor values or total-harmonic-distortion measurement are preserved. The plots show operation near the supply rails, but they do not establish an exact linear-swing or distortion boundary.

A clear boundary around the result.

Requested evidence	Archive status
Nominal loop gain / stability	Present: annotated loop plot, screenshot result table, final STB record.
Common-mode sweep	Present: f_u and phase margin versus VCM.
DC transfer / load current	Present: 10 kohm baseline and 2 kohm stress sweeps.
Quiescent output bias	Present: annotated M64 / M67 operating points.
Unity-step response / slew	Present: step waveform and positive slew cursors; numerical SR- absent.
Gain +2 / gain -1 sine response	Present: 10 kHz waveforms; swing is approximate.
Complete schematic / sizing table	Partial: local operating-point views; no consolidated high-resolution schematic or verified sizing table.
Hand-analysis comparison	Not fully preserved for final implementation.
PVT / mismatch / layout	Not established by the supplied evidence.

The nominal design point meets the recorded GBW and PM thresholds with modest margin. The next engineering step is to create one reproducible final test state, export the complete schematic and sizing table, measure negative slew and exact output swing, and assess robustness beyond nominal conditions.

No noise, CMRR, PSRR, THD, Monte Carlo, or post-layout result is claimed. Report values are linked to their own saved run or screenshot instead of combining multiple design snapshots into one sign-off table.

Evidence index.

Paths are relative to 'design challenge2'. The supplied three-page design brief defines the topology and tests. The final nominal record is montecelli / folded_ota / adexl / results / data / Interactive.184.log and .rdb: GBW = 10,108,286.739 Hz; PM = 60.5839063 degrees.

SOURCE SCREENSHOTS

- 01. screenshots / Screenshot 2026-09-10 172935.png
- 02. screenshots / Screenshot 2026-09-10 172358.png
- 03. screenshots / Screenshot 2026-09-10 172407.png
- 04. screenshots / Screenshot 2026-09-10 172628.png
- 05. screenshots / Screenshot 2026-09-10 173635.png
- 06. screenshots / Screenshot 2026-09-10 173701.png
- 07. screenshots / Screenshot 2026-09-10 174020.png
- 08. screenshots / Screenshot 2026-09-10 174102.png
- 09. screenshots / Screenshot 2026-09-10 174037.png
- 10. screenshots / Screenshot 2026-09-10 174046.png
- 11. screenshots / Screenshot 2026-09-10 174552.png
- 12. screenshots / Screenshot 2026-09-10 174622.png
- 13. screenshots / Screenshot 2026-09-10 174603.png
- 14. screenshots / Screenshot 2026-09-10 174612.png
- 15. screenshots / Screenshot 2026-09-10 175449.png
- 16. screenshots / Screenshot 2026-09-10 175435.png
- 17. screenshots / Screenshot 2026-09-10 182432.png
- 18. screenshots / Screenshot 2026-09-10 182734.png

Source screenshots retain their original labels and measurement annotations. Approximate values in this report are derived from visible axes; original waveform data is required for greater precision.